ModelIt!

Student Activity Pack

How Do Stars Forge Every Atom in Your Body?

Modeling Stellar Nucleosynthesis and the Origin of the Elements

Grade 10L2

NGSS HS-ESS1-3, HS-PS1-5

Energy and Matter: Flows, Cycles, and Conservation

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 Where do you think the atoms in your body originally came from?

2 What do you know about how stars produce energy?

3 Draw a diagram showing what you think happens inside a star during its lifetime.

4 Why do you think there are so many different elements on the periodic table instead of just one or two?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Stellar Core Temperature | Hydrogen Fusion Rate | Helium Accumulation | Gravitational Pressure | Radiation Output | Element Production | Star Lifecycle Stage

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Sun-Like Star

Moderate mass, standard fusion

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Massive Star

High mass, rapid burning

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Red Dwarf

Low mass, slow burning

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Stellar Mass — The total mass of the star, which is the master variable determining core temperature, fusion rate, element production capacity, lifespan, and ultimate fate

Neutrino Emission — The flood of nearly massless particles produced during fusion and especially during core collapse, which carry away energy and signal the star's internal state

Metallicity — The proportion of elements heavier than helium in the star's composition, inherited from previous stellar generations and affecting opacity, fusion rates, and stellar structure

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Every atom in your body was forged inside a star that exploded before our Sun was born — how did stars create the entire periodic table? Use evidence from your simulation to support your answers.

1. **How does your model demonstrate that a star's mass determines its entire life story — from birth to death?**

2. **Why is iron the 'dead end' of stellar fusion, and what does this mean for how elements heavier than iron are created?**

3. **What does your model reveal about the timescales involved — why does hydrogen burning last billions of years while silicon burning lasts only days?**

4. **How does the concept of hydrostatic equilibrium (gravitational pressure vs. radiation output) explain both stellar stability and stellar death?**

5. **What are the limitations of modeling stellar evolution with only seven components when real stars involve magnetohydrodynamics, rotation, and mass loss?**

8 STEM Challenge: Design a Stellar Evolution Visualization Tool

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YOUR ENGINEERING MISSION

Design an interactive visualization that shows how stars of different masses produce different elements and how those elements end up in planets and living organisms.

THE SCENARIO

A science museum is creating a new exhibit on the origin of elements. They need an interactive display that lets visitors select a star mass and watch the entire lifecycle unfold — from birth through element production to death. Your team has been hired to design this visualization using your model data.

Engineering Design Thinking

1. WHAT VISUAL METAPHOR WILL MAKE NUCLEAR FUSION INTUITIVE FOR A GENERAL AUDIENCE?

2. HOW WILL YOU SHOW THE ONION-LAYER STRUCTURE OF A MASSIVE STAR'S CORE?

3. WHAT DATA FROM YOUR SIMULATIONS WILL DRIVE THE VISUALIZATION'S BEHAVIOR?

4. HOW WILL YOU HANDLE THE DRAMATIC DIFFERENCE IN TIMESCALES — BILLIONS OF YEARS OF HYDROGEN BURNING VERSUS SECONDS OF CORE COLLAPSE?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Astrophysicists study the physical processes governing stars, galaxies, and the universe. They work at universities, NASA, national observatories, and research institutions, earning $70,000–$150,000/year. Nuclear Physicists who study fusion reactions earn $80,000–$170,000/year.